

Variable and constraint reduction techniques for the temporal bin packing problem with fire-ups

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Abstract

The aim of this letter is to design and computationally test several improvements for the compact integer linear programming (ILP) formulations of the temporal bin packing problem with fire-ups (TBPP-FU). This problem is a challenging generalization of the classical bin packing problem in which the items, interpreted as jobs of given weight, are active only during an associated time window. The TBPP-FU objective function asks for the minimization of the weighted sum of the number of bins, viewed as servers of given capacity, to execute all the jobs and the total number of fire-ups. The fire-ups count the number of times the servers are activated due to the presence of assigned active jobs. Our contributions are effective procedures to reduce the number of variables and constraints of the ILP formulations proposed in the literature as well as the introduction of new valid inequalities. By extensive computational tests we show that substantial improvements can be achieved and several instances from the literature can be solved to proven optimality for the first time.

Keywords: Cutting and Packing · Temporal Bin Packing Problem · Fire-Ups · Integer Linear Programming

MSC: 90-10 , 90C10 , 90C90

1 Introduction

The *temporal bin packing problem* (TBPP) introduces a temporal dimension to the classical *bin packing problem* (BPP), see [11, 18, 20], by associating to the items time windows in which they are active. Formally, given a set of *items* (or *jobs*), the TBPP asks for determining the minimum number of *bins* (or *servers*) to execute all jobs. Each job is characterized by a *size* (or *resource demand*) and a *lifespan* (time window in which the job is active), a jobs-to-servers assignment is feasible if and only if the capacity of the servers is respected at any instant of time. The TBPP is a challenging problem with a high practical and theoretical interest which has been recently introduced in the literature, see [9, 10]. It belongs to the rich family of cutting and packing problems, object of intense research in the last decades. The TBPP can also be interpreted as a high-dimensional vector packing problem [7, 19] and it partially shares the mathematical structures of two-dimensional packing problems, where the time can be seen as one of the dimensions and the other one is related to the capacity of the servers. We also refer the interested reader to [13, 16] for the strip packing versions of these problems.

Another problem closely related to the TBPP is the *temporal knapsack problem* (TKP). The TKP asks for determining a maximum-profit subset of jobs (also active for a given time window) which can be executed by a single server, see [2, 4]. As far as the exact approaches to solve the TKP to proven optimality are concerned, this problem is usually tackled by a Dantzig-Wolfe reformulation and branch-and-price algorithms [5, 6] or by dynamic programming algorithms [8].

The TBPP is introduced in an application-oriented article, see [9], dealing with efficient workload server management in data centers – one of the key challenges in light of the ever-growing energy demand of the IT industry [1, 12, 14]. As far as the TBPP solution methods are concerned, we refer the reader to [10], an article which presents several heuristic and exact approaches. The state-of-the-art exact algorithm for the TBPP is a branch-and-price algorithm, which exploits in preprocessing a plethora of heuristic algorithms (see [10] for further details).

In the TBPP, the quality of a jobs-to-servers assignment is evaluated solely by the number of servers in use. However, several recent publications point out that the specific operating mode of the servers is also crucial. This means, in particular, that an inactive server can be temporarily put into a sleep mode (or can be completely shut down) for the purpose of saving energy. In this case the server must then be turned on again, if required. A server restart is called a *fire-up*, see also Fig. 1 for a graphical representation, and – from the perspective of energy-efficiency – this naturally leads to a second optimization goal, introduced in [3]. The authors of this article propose two compact ILP formulations (denoted M1 and M2) taking into consideration two different objectives: *i*) the number of servers and *ii*) the number of fire-ups. This new objective function is a weighted-sum, using the input parameter $\gamma > 0$ to scale the contribution of the fire-ups. In this way, a multi-objective variant of the TBPP has been proposed in [3], hereinafter referred to as the *temporal bin packing problem with fire-ups* (TBPP-FU). The models M1 and M2 of [3] are based on the classic Kantorovich-type structure for the BPP, see [15], and the experiments show that the proposed exact solution method based on these models is very challenging from a computational point of view. In order to quickly find good-quality heuristic solutions, a *constructive look-ahead heuristic* (CLH), together with an advanced recovery algorithm based on mathematical programming techniques, is also presented in [3].

In this article, an interesting insight of the multi-objective function is also shown: for rather small values of γ , i.e., for $\gamma \leq 1/n$ (where n is the number of jobs), it is possible to considerably reduce the size of the ILP formulations by exploiting the information given by the heuristic solutions. However, the important question on how to reduce the models in the general case of arbitrary values of γ is still an open problem which we tackle in this letter.

Very recently, in [17] several methods for improving the TBPP-FU ILP models are proposed. These techniques generate substantial performance gains, including also the possibility of efficiently handling larger values of γ . In addition, a third formulation (called M3) is proposed which is not based on the classical job-to-server assignment variables. More in details, M3 is based on a job-to-job relation and it allows to a priori discard some infeasible solutions by removing a set of variables. As a result, model M3 is a more compact ILP formulation (compared to M1 and M2) which is particularly effective for fast calculations. However, from an overall point of view, the tests in [17] show that the “optimized” version of M1 remains on average the state-of-the-art approach in terms of instances solved to proven optimality.

In this article, we aim to enrich and complete the structural analysis of the compact ILP formulations for the TBPP-FU. To this end, we present new methods to improve the ILP models which lead to improved numerical results. More in details, we show how to transfer parts of the favorable properties of M3 to M1 and M2, obtaining, in this manner, very robust and more powerful compact formulations. The main contributions of our investigation are:

- We optimize and reduce the set of constraints of M1 by removing redundancies and tightening some conditions.
- We present a new class of valid inequalities for M1 and M2 mimicking the inherent property of M3 to avoid forbidden item pairs.

- We theoretically show how the heuristic-based information can be used for arbitrary choices of γ , generalizing the results of the literature.

The remainder of this letter is structured as follows: In the next section we introduce the notation and the M1 and M2 formulations from the literature. In the core Sect. 3, we present the new reduction procedures and the new family of valid inequalities. Finally, extensive computational tests are reported in Sect. 4, aiming at computationally evaluating the beneficial effect of the newly proposed techniques.

2 Preliminaries and Basic Models

In this section, we present the most important terminologies and notations, as well as a brief overview of the current ILP formulations from the literature. Let us consider $n \in \mathbb{N}$ *items (jobs)*, each of which possessing a *resource demand (item size)* $c_i \in \mathbb{N}$ that is active only in the interval $[s_i, e_i]$ formed by the *starting time* $s_i \in \mathbb{N}$ and the *ending time* $e_i \in \mathbb{N}$, $i \in I := \{1, \dots, n\}$. Then, these items have to be assigned to *bins (servers)* of capacity $C \in \mathbb{N}$ so that a weighted sum consisting of (i) the number of servers required and (ii) the number of fire-ups is minimized, where the second term is scaled by a weighting parameter $\gamma > 0$. Let $K := \{1, \dots, n\}$ denote the set of servers and let $T := \bigcup_{i \in I} \{s_i, e_i\}$ and $T_S := \bigcup_{i \in I} \{s_i\}$ collect the relevant instants of (starting) times. Moreover, we assume the items to be ordered with respect to non-decreasing starting times (where ties are broken arbitrarily), and we use $I_t := \{i \in I \mid t \in [s_i, e_i]\}$ to indicate the active jobs at time $t \in T$.

Example 2.1. Let us consider an instance with servers of capacity $C = 2$ and $n = 5$ items having resource demands $c = (1, 1, 1, 1, 1)$, starting times $s = (1, 1, 5, 7, 7)$, and ending times $e = (2, 11, 15, 16, 8)$. Then, for $\gamma = 1$, an optimal solution requires two servers and three fire-ups as depicted in Fig. 1.

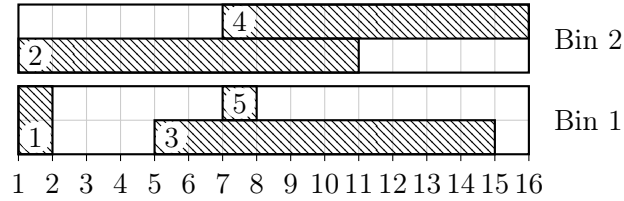


Figure 1: An optimal assignment for $\gamma = 1$.

In total, three different compact models for the TBPP-FU have so far been formulated in the literature. To reiterate, the original publication [3] contained two ILP formulations (called M1 and M2) which were based on classic job-to-server assignment variables. Some first reduction methods together with a new third exact approach (called M3) have been presented in [17], all of which contributed to significantly better numerical results. Here we only present the basic versions of the two models from [3] in some detail, since our new reductions either specifically address the set of servers to be modeled (which, however, is not explicitly contained in M3) or exclude combinations of items (which is automatically done in M3 through job-to-job coupling). Moreover, among the three approaches considered, M1 turned out to perform best (on average), so that its further investigation seems to be particularly interesting.

To state M1, let us consider the following four types of variables:

- We have $z_k = 1$ if and only if server $k \in K$ is used.
- We set $x_{ik} = 1$ if and only if job $i \in I$ runs on server $k \in K$.
- We use $y_{tk} = 1$ if and only if server $k \in K$ is active at time $t \in T$.

- We have $w_{tk} = 1$ if and only if server $k \in K$ is activated at time $t \in T_S$.

Consequently, the z -variables measure the number of servers required whereas the w -variables are responsible for counting the fire-ups. The original version of M1, as presented in [3], then reads as follows

Model 1 (M1)

$$\begin{aligned}
& \min \sum_{k \in K} \left(z_k + \gamma \sum_{t \in T_S} w_{tk} \right) \\
\text{s.t.} \quad & y_{tk} \leq \sum_{i \in I_t} c_i x_{ik} \leq y_{tk} C, & k \in K, t \in T, & (1) \\
& \sum_{k \in K} x_{ik} = 1, & i \in I, & (2) \\
& x_{ik} \leq y_{s_i, k}, & i \in I, k \in K, & (3) \\
& y_{tk} \leq z_k, & k \in K, t \in T, & (4) \\
& y_{tk} - y_{t-1, k} \leq w_{tk}, & k \in K, t \in T_S, & (5) \\
& x_{ik} \in \{0, 1\}, & i \in I, k \in K, & (6) \\
& y_{tk} \in \{0, 1\}, & k \in K, t \in T, & (7) \\
& w_{tk} \in \{0, 1\}, & k \in K, t \in T_S, & (8) \\
& z_k \in \{0, 1\}, & k \in K. & (9)
\end{aligned}$$

The objective function minimizes a weighted sum of the aforementioned criteria. Constraints (1) make sure that the capacity of an active server is respected (right hand side), and that an empty server is deactivated (left hand side). Conditions (2) demand that any job is assigned exactly once, while linking the different types of variables is done by Restrictions (3)-(5). In particular, (5) is responsible for recognizing a fire-up in precisely those cases where the considered server is currently active, but was inactive at the predecesing instant of time (indicated by $t - 1$ in a symbolic way).

The second formulation M2 appearing in [3] addresses the temporal aspect of the problem in a less explicit way, e.g., by not making use of the y -variables measuring the servers' activity. Instead, however, the sets $\delta_i := \{j < i \mid s_i < e_j\}$ and $\delta_i^+ := \{j < i \mid s_i \leq e_j\}$, $i \in I$, gathering jobs being active at s_i (see δ_i and δ_i^+) or just having finished at s_i (only δ_i^+) are required. With these ingredients, M2 is given by:

Model 2 (M2)

$$\begin{aligned}
& \min \sum_{k \in K} \left(z_k + \gamma \sum_{t \in T_S} w_{tk} \right) \\
\text{s.t.} \quad & \sum_{j \in \delta_i} c_j x_{jk} + c_i x_{ik} \leq C z_k, & i \in I, k \in K & (10) \\
& \sum_{k \in K} x_{ik} = 1, & i \in I, & (11) \\
& x_{ik} \leq z_k, & i \in I, k \in K, & (12) \\
& w_{s_i, k} \geq x_{ik} - \sum_{j \in \delta_i^+} x_{jk}, & i \in I, k \in K, & (13) \\
& x_{ik} \in \{0, 1\}, & i \in I, k \in K, & (14) \\
& w_{tk} \in \{0, 1\}, & t \in T_S, k \in K, & (15) \\
& z_k \in \{0, 1\}, & k \in K. & (16)
\end{aligned}$$

The objective function and Conditions (11) already appeared in exactly the same way in M1, whereas Constraints (10) again ensure that the server capacity is respected. Restrictions (12)-(13) are responsible for linking the different types of variables. In particular, Conditions (13) state that a fire-up at time s_i has to be perceived on server k , if item i , but none of the items from δ_i^+ , has been placed on it.

Besides being relatively large in size, the original versions of M1 and M2 possess some structural drawbacks: (i) The solution space is highly symmetric. (ii) The LP relaxation is rather poor. (iii) The set K is much larger than required in an optimal solution. Note that the first two problems are mainly related to the Kantorovich-type structure of the models and were already been successfully addressed in parts in [17]. Without going into too much detail (for this, we refer the reader to the contributions of [17]), this was done primarily by:

- (R1) Renumbering the servers so that only index pairs (i, k) from the set $\Delta := \{(i, k) \mid k \leq i\}$ have to be considered. This also made it possible to move to server-dependent time sets $T(k)$ and $T_S(k)$, which helped to also save some of the y - and w -variables.
- (R2) Introducing valid inequalities $z_k \leq \sum_{t \in T_S(k)} w_{tk}$ for any $k \in K$ implying at least one fire-up on every server. In particular, this prevents the two variable types in the objective function from becoming arbitrarily small independently of each other.
- (R3) Lifting the item sizes c_i , $i \in I$, to possibly tighten Conditions (1) and (10).

Furthermore, by additional minor reductions (referred to as (R4) and (R5) in [17]), some redundancies within the set of constraints could be eliminated in M2. Based on numerical tests, it could be shown that the listed techniques lead to significant improvements of the compact models, with (R1) and (R2) proving to be particularly valuable for the benchmark instances considered. This was due mainly to the fact that the size of the models could be reduced by roughly 50% (both in terms of variables and constraints), but also to significantly better LP bounds. For this reason, it seems worthwhile to us to explore further reduction possibilities and present them in the next section, thus arriving at the somewhat “best possible” version of the compact formulations M1 and M2.

3 New Reduction Methods

The new reduction methods to be proposed in this section can be roughly divided into three categories: (a) “optimizing” the set of constraints, (b) adding new valid inequalities, and (c) a heuristic-based reduction of the model size. While item (a) is specific to the y -variables and Constraints (1) from M1, (b) and (c) can be applied to both models. For this reason, we will discuss all these methods using M1, but also briefly point out how they might need to be modified for M2.

3.1 Reduction (a)

We first observe that after having applied Reduction (R1), Constraints (1) from M1 only need to be formulated for the time steps $t \in T(k)$ relevant on server $k \in K$. However, we do not require the whole inequality chain

$$y_{tk} \leq \sum_{i \in I_t: (i, k) \in \Delta} c_i x_{ik} \leq y_{tk} C$$

for each $t \in T(k)$, since either of the two parts has a very specific purpose. To be more precise, the left hand side is only important to perceive the deactivation of a server, while the right hand side helps to recognize an active server. Hence, it suffices to state the first inequality for ending times $t \in T_E(k) = \bigcup_{i \geq k} \{e_i\}$ (on server k) only, whereas the second inequality needs to be formulated just for

the server-dependent starting times $t \in T_S(k) = \bigcup_{i \geq k} \{s_i\}$. In the latter case, we can even go a step further by noting that only the non-dominated starting times $T_S^{nd}(k)$ have to be considered. Observe that a starting time t_1 is called *dominated* if it is directly followed by another starting time t_2 (so that every job that is active at t_1 is still active at t_2), see also [10, 17]. After we have accordingly broken the chain of inequalities into two parts, we can strengthen the first part by moving from $y_{tk} \leq \sum_{i \in I_t: (i,k) \in \Delta} c_i x_{ik}$ to $y_{tk} \leq \sum_{i \in I_t: (i,k) \in \Delta} x_{ik}$. Obviously, the latter better conveys the important message that an empty server cannot be active and an active server needs to have at least one item running on it. Overall, Reduction (a) helps to save some redundant conditions appearing in (1), while other conditions of that type are even tightened.

3.2 Reduction (b)

One of the major advantages of Model M3 from [17] consisted of the fact that the set of variables to be considered was very small due to the elimination of *forbidden item pairs*. By that, we mean that any pair of jobs appearing in $F := \{(i, j) \mid [s_i, e_i) \cap [s_j, e_j) \neq \emptyset, c_i + c_j > C\}$ cannot be executed on the same server. While this could be incorporated directly into the model generation of M3 due to the job-to-job coupling, additional valid inequalities are needed for M1 and M2 to avoid such pairs. However, the naive way of just demanding $x_{ik} + x_{jk} \leq z_k$ for any $k \in K$ and any $(i, j) \in F$ with $(i, k), (j, k) \in \Delta$ normally leads to a very large amount of further constraints (up to $\mathcal{O}(n^3)$ of them in the worst case). For this reason, here we propose a more sophisticated strategy that produces fewer and at the same time (partly) better such inequalities. To this end, note that for a fixed server k it would be sufficient to collect (a reasonable subset of) the maximal cliques of the *incompatibility graph* $\mathcal{G}(k) = (I(k), F)$ formed by the relation F on the set $I(k) := \{i \in I \mid i \geq k\}$ of items that can be assigned to server k . Then, for any such clique $\mathcal{C}$ (related to server k) with $|\mathcal{C}| \geq 2$, the condition

$$\sum_{i \in \mathcal{C}} x_{ik} \leq z_k \tag{17}$$

has to be added. For M1, we recommend to replace z_k on the right hand side of this inequality by y_{tk} , where t is the last starting time of the items from $\mathcal{C}$ (that is, one specific instant of time where all these items are active). Thanks to the coupling conditions (4) from M1, this will impose an even stronger constraint.

To find the maximal cliques, we propose the following strategy, starting with $k = 1$ (so that $I(k) = I$ holds):

- i) Find the maximal cliques of the subgraph (of $\mathcal{G}(1)$) formed only by the items having $c_i > C/2$.
- ii) For any fixed item $i \in I$ with $c_i \leq C/2$: consider the subgraph formed by the items j with $c_j > C - c_i$ that are adjacent to i in $\mathcal{G}(1)$. When we add $\{i\}$ to a maximal clique of this subgraph, then we end up with a maximal clique of $\mathcal{G}(1)$.

By these two steps, we will find all maximal cliques of $\mathcal{G}(1)$ thanks to the following result:

Lemma 3.1. *Let $k \in K$ be fixed. Then, a maximal clique of $\mathcal{G}(k)$ can have at most one item of size $\leq C/2$.*

Proof. Indeed, if there were two such items in the clique, we would not have an edge between them which gives the contradiction. $\square$

For any remaining server $k \geq 2$, the maximal cliques of $\mathcal{G}(k)$ (having size $|\mathcal{C}| \geq 2$) can be iteratively obtained from the information of the previous step $k - 1$ by deleting the item $i = k - 1$ from any maximal clique of $\mathcal{G}(k - 1)$ and applying cardinality and dominance tests to the obtained subsets of items.

3.3 Reduction (c)

The techniques presented so far (in Sect. 2 and 3) have not yet contributed to the reduction of the quantity $|K|$, which has a decisive influence on the model size as, for instance, the index k appears in any of the four variable types of M1. To deal with this challenge, which was already established as item (iii) in Sect. 2, appropriate heuristic information can be applied. However, from a theoretical point of view, the latter has only been successfully achieved for very small choices of γ :

Theorem 3.2 ([3]). *Let $\gamma \leq 1/n$. Then, the number of servers required for the TBPP-FU is equal to the number of servers in an optimal solution to the TBPP.*

The previous result allows for either computing the optimal size of K beforehand by solving a somewhat easier auxiliary problem (that is, the TBPP) or to at least limit the number of servers to any value obtained by a heuristic solution. However, as also reported in [3, Example 2.2], this result does not hold for larger choices of γ , so that reducing the size of K cannot be performed in the majority of the cases. To tackle this issue the following result presents an easy way of using heuristic information in the general case, too.

Theorem 3.3. *Let z_{heu} be the objective value obtained by any heuristic for the TBPP-FU. Then, the number of servers required in an optimal solution is at most $k^* := \lfloor z_{heu}/(1 + \gamma) \rfloor$.*

Proof. If the claim was wrong we would need at least $k^* + 1$ servers in an optimal solution. Since every server is switched-on at least once, this would lead to an objective value of at least

$$(k^* + 1) + \gamma \cdot (k^* + 1) = (1 + \gamma) \cdot (k^* + 1) > z_{heu}$$

giving the contradiction because the heuristic would have to be better than the optimal solution. $\square$

This result allows us to replace the set $K = \{1, \dots, n\}$ at all positions in M1 and M2 with an appropriately defined and greatly reduced set $K^* := \{1, \dots, k^*\}$. For our investigations, we will use the constructive look-ahead heuristic (CLH) described in [3, Sect. 3], but in a slightly more exploratory way. Before explaining the precise meaning of this intention, let us briefly collect the main idea of that heuristic: As stated in Sect. 2, we start by an item list ordered with respect to non-decreasing starting times s_i (where ties are broken in an arbitrary way) and process the items one by one. Moreover, we require a *look-ahead parameter* $q \in \mathbb{N}$ indicating the number of future items to be taken into account when making the current decision. In a specific iteration, we consider a fixed item $i \in I$ and assign it to every open server that is able to accommodate it, and (as another alternative) also to a new empty server. By that, we obtain various different assignments $\mathcal{A}_1, \dots, \mathcal{A}_p$. Now, we add the next q items to any of these assignments in a best-fit fashion, leading to the extended assignments $\tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_p$. Finally, we compute the corresponding objective values (i.e., the weighted sum of servers and fire-ups) and place item i to that bin whose extended assignment led to the lowest objective value.

Since in [3] the parameter $q = 3$ was used without compelling justification, we will first preface our actual test calculations in the next section with a somewhat more detailed consideration of the CLH algorithm.

4 Computational Tests

For our numerical calculations, we coded the above approaches in Python (version 3.9.2) and used its Gurobi (version 9.1.1) interface to solve the resulting ILP formulations with default settings and a time limit of 30 minutes per instance. All the experiments were run on an AMD A10-5800K processor with 16 GB RAM. Due to its novelty, the TBPP-FU has not yet been able to leave a large scientific footprint in the relevant literature, so that only one set of benchmark instances has been specifically designed for the problem under consideration, see [3]. In that publication, the authors propose 160 instances formed by 32 groups of five instances each, all sharing the same capacity $C = 100$. Apart from that, any group is determined by four indicators:

- number of items: $n \in \{50, 100, 150, 200\}$,
- time horizon: dense or relaxed (indicated by a maximum starting time $\bar{s} \in \{n, 1.2n\}$),
- job duration $d_i := e_i - s_i$: short or long (represented by ' d_S ' and ' d_L ' and indicated by uniformly distributed integers $d_i \in [10, 30]$ and $d_i \in [20, 60]$, respectively)
- resource consumption c_i : low or high (represented by ' c_L ' and ' c_H ' and indicated by uniformly distributed integers $c_i \in [25, 50]$ and $c_i \in [25, 75]$, respectively).

Even though the total number of instances appears to be relatively small, they can be considered very suitable for numerical test calculations due to their difficulty, especially because only 63 of them could be solved in the original publication [3]. Also the improvements discussed in [17] could increase this number to only just over 50% (that is, 85 out of 160), so that their solution still represents a serious challenge from today's point of view.

In a first experiment, we study the influence of the look-ahead parameter q on the performance of the CLH approach from [3]. For this purpose, we exemplarily consider the instances with $n \in \{100, 200\}$ items and refer to the results in Tab. 1. Based on this data, one can see the rough trend that a deeper look into the future (that is, a larger value of q) basically leads to a reduction in the heuristic value. However, this is by no means a strictly monotonous relationship, because for two different choices of q the obtained assignments will be considerably different, in general. Consequently, although the tabulated data give a very consistent picture in that large values of q are to be preferred, there is no single universally best choice of that parameter.

Remark 4.1. To better evaluate these data, column LB in Tab. 1 contains the average rounded-up LP values of $M1^*$, i.e., a lower bound for the integer optimal value. By that, we see for instance that the average difference between the heuristic and the optimal value is bounded by roughly 27% for the hard instances with $n = 200$ items, but for a few constellations (especially with c_H) it is also (considerably) larger since it is much harder to obtain a dense heuristic packing in these cases.

As for the computational efforts, it is important to note that all the heuristic values can be determined very quickly, meaning that for many scenarios (n, q) the heuristic solution is available in less than one second. Even checking all the look-ahead parameters q (mentioned in Tab. 1) for an instance with $n = 200$ jobs and then deciding on the best result (see column 'best' in Tab. 1) takes only about 17 seconds on average, which is quite acceptable when measured against the time limit of 30 minutes permitted for the exact solution of these rather challenging instances. For this reason, and considering that our intention is to present a preferably maximally reduced compact formulation, we will always choose the best heuristic value to define k^* (that is, the number of initialized servers) appearing in Theorem 3.3. We note, however, that one could alternatively agree on a compromise between computational effort and quality of the heuristic solution and always use a fixed value of the look-ahead parameter

(say $q = 20$), since already this leads to a significant improvement (e.g., on average about 15% better heuristic values for $n = 200$) compared to the relatively arbitrary choice of $q = 3$ from [3], without noticeably increasing the time required. Either way, as the cardinality of K strongly influences the number of variables and constraints, very powerful reductions in terms of the model size can be expected.

n	$\bar{s}$	d_i	c_i	1	2	3	5	10	20	n/4	n/2	n	best	LB
100	100	d_S	c_L	30.0	30.2	29.2	29.4	27.6	27.4	26.8	25.8	27.0	25.6	22.4
			c_H	48.4	46.4	45.0	43.0	40.4	39.2	39.4	40.4	39.4	38.2	33.6
		d_L	c_L	40.6	40.4	41.0	39.8	38.4	37.8	38.4	38.0	37.8	37.4	34.4
			c_H	66.2	64.4	62.6	60.6	58.6	55.2	56.0	55.0	54.4	54.4	49.2
	120	d_S	c_L	28.6	27.8	26.6	25.6	24.2	24.2	24.8	24.0	25.0	23.2	20.0
			c_H	45.6	42.4	41.6	39.8	39.0	38.2	38.4	39.6	39.4	37.2	27.2
		d_L	c_L	36.2	36.2	36.0	34.6	34.2	33.6	33.8	32.8	33.2	32.6	30.8
			c_H	60.0	59.2	57.6	55.2	53.0	55.2	51.6	51.6	51.0	50.2	44.4
	Average				44.5	43.4	42.5	41.0	39.4	38.9	38.7	38.4	38.4	37.4
200	200	d_S	c_L	40.4	39.4	38.4	37.0	34.0	32.8	33.4	34.6	34.6	32.0	24.4
			c_H	69.4	67.8	65.4	60.4	55.8	53.0	53.8	57.6	58.4	51.8	31.6
		d_L	c_L	50.0	49.2	47.4	47.8	45.8	44.0	42.8	42.4	43.2	42.0	38.0
			c_H	89.8	86.0	83.6	78.8	72.6	69.8	68.0	67.8	69.4	66.6	53.6
	240	d_S	c_L	39.6	38.2	38.4	36.2	34.8	30.6	32.0	33.0	34.2	30.2	21.2
			c_H	76.6	71.6	67.6	64.4	60.4	59.0	59.2	61.4	61.2	58.0	30.6
		d_L	c_L	43.4	42.6	41.0	42.6	39.6	39.0	37.4	38.0	37.8	36.8	32.4
			c_H	84.0	81.2	77.6	71.8	67.6	62.0	62.6	62.8	65.6	61.2	46.0
	Average				61.7	59.5	57.4	54.9	51.3	48.8	48.6	49.7	50.6	47.3

Table 1: Heuristic values for different choices of the look-ahead parameter q . The best average value for each subset is printed in bold. The column 'best' refers to the average over the best value obtained per instance from the considered subset. The column 'LB' provides the (average) rounded-up LP value of M1* to enable a rough evaluation of the heuristic solution.

n	$\bar{s}$	d_i	c_i	n_{var}					n_{con}							
				M1	M1*	M2	M2*	M3	M1	M1*	M2	M2*	M3			
100	100	d_S	c_L	13.5	2.9	8.4	2.0	5.2	24.0	3.6	13.8	3.2	13.4			
			c_H	13.5	4.2	8.3	2.8	4.4	24.1	6.7	14.1	6.2	9.8			
		d_L	c_L	14.4	4.1	8.4	2.8	5.2	26.6	4.6	13.7	4.3	13.4			
			c_H	14.3	5.6	8.3	3.8	3.8	26.4	9.9	13.7	9.4	9.0			
	120	d_S	c_L	14.2	2.8	8.6	1.8	5.2	25.5	3.4	13.9	2.9	13.6			
			c_H	14.2	4.4	8.6	2.9	4.5	25.6	7.1	13.7	6.2	10.0			
		d_L	c_L	14.7	3.8	8.6	2.5	5.2	27.2	4.4	13.8	3.9	13.6			
			c_H	15.0	5.7	8.7	3.7	3.9	27.7	9.9	13.5	9.1	9.2			
			200	200	d_S	c_L	52.5	7.6	33.4	5.0	20.3	91.0	9.6	54.9	8.3	53.4
						c_H	51.8	11.8	33.1	7.9	18.7	89.8	20.9	56.2	19.0	41.0
d_L	c_L	54.8			9.8	33.4	6.5	20.3	97.8	11.9	55.3	10.5	53.4			
	c_H	53.3			14.7	32.9	9.9	17.2	94.3	28.6	54.9	26.7	39.2			
240	d_S	c_L		55.1	7.5	34.0	4.8	20.3	97.6	9.3	54.2	7.7	53.9			
		c_H		55.9	14.4	34.4	9.2	18.9	99.3	23.4	54.2	20.2	42.3			
	d_L	c_L		56.9	9.1	34.2	5.9	20.3	102.6	11.0	54.7	9.2	54.2			
		c_H		56.5	14.7	34.1	9.5	17.7	101.7	26.3	55.0	23.4	40.7			
		Average (for $50 \leq n \leq 200$)				26.0	6.1	15.9	4.0	8.9	46.5	9.4	25.8	8.4	21.9	

Table 2: An overview of the numbers of variables n_{var} and constraints n_{con} (all of which represented in units of 10^3). M1, M2, and M3 are the versions from the literature, see [17], whereas M1* and M2* contain the ideas from Sect. 3.

For the sake of exposition, we take a closer look at the associated numbers in Tab. 2. Due to space limitations, we again consider only the subset of instances that was also used in Tab. 1, but finally we also report the average results over all 160 instances in the last row of Tab. 2 to allow for a better overall picture. In addition, we also include the results of M3 from [17], as this is the best formulation known in the literature so far in terms of model size. Compared to that approach, we notice that the

n	$\bar{s}$	d_i	c_i	M1		M1*		M2		M2*		M3			
				t	opt	t	opt	t	opt	t	opt	t	opt		
50	50	d_S	c_L	7.4	(5)	0.9	(5)	6.3	(5)	1.5	(5)	3.6	(5)		
			c_H	8.6	(5)	1.5	(5)	2.8	(5)	4.3	(5)	3.7	(5)		
		d_L	c_L	367.8	(4)	254.3	(5)	71.9	(5)	360.3	(4)	17.0	(5)		
			c_H	12.3	(5)	0.9	(5)	5.0	(5)	0.5	(5)	0.2	(5)		
	60	d_S	c_L	14.0	(5)	0.9	(5)	3.3	(5)	0.5	(5)	4.6	(5)		
			c_H	37.8	(5)	4.4	(5)	363.1	(4)	3.9	(5)	3.2	(5)		
		d_L	c_L	251.2	(5)	5.5	(5)	5.6	(5)	5.0	(5)	1.1	(5)		
			c_H	5.7	(5)	0.9	(5)	2.2	(5)	0.9	(5)	0.3	(5)		
		Average (Sum)				88.1	(39)	33.7	(40)	57.5	(39)	47.1	(39)	4.2	(40)
		100	100	d_S	c_L	22.1	(5)	1.7	(5)	4.2	(5)	0.4	(5)	46.8	(5)
c_H	738.6				(4)	376.3	(5)	956.7	(3)	735.2	(3)	1128.7	(3)		
d_L	c_L			1457.9	(1)	1444.9	(1)	1800.0	(0)	1442.7	(1)	1447.4	(1)		
	c_H			1500.4	(1)	1458.7	(1)	1555.6	(1)	1452.0	(1)	1443.4	(1)		
120	d_S		c_L	91.2	(5)	8.3	(5)	152.7	(5)	51.4	(5)	381.8	(4)		
			c_H	1042.8	(3)	877.1	(3)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
	d_L		c_L	850.4	(3)	517.0	(4)	803.8	(3)	377.2	(4)	635.2	(4)		
			c_H	1481.0	(1)	861.1	(3)	1329.9	(2)	1082.8	(2)	733.3	(3)		
	Average (Sum)				898.0	(23)	693.1	(27)	1050.4	(19)	867.7	(21)	952.1	(21)	
	150		150	d_S	c_L	462.4	(4)	69.3	(5)	404.5	(4)	198.4	(5)	601.1	(4)
c_H		1800.0			(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
d_L		c_L		1502.3	(1)	1455.7	(1)	1800.0	(0)	1440.4	(1)	1726.6	(1)		
		c_H		1517.8	(1)	1504.5	(1)	1800.0	(0)	1448.4	(1)	1628.7	(1)		
180		d_S	c_L	93.7	(5)	4.4	(5)	377.7	(4)	24.4	(5)	634.1	(4)		
			c_H	1682.6	(2)	1200.4	(3)	1800.0	(0)	1709.1	(1)	1800.0	(0)		
		d_L	c_L	1800.0	(0)	1108.5	(2)	1620.0	(1)	1184.1	(2)	1800.0	(0)		
			c_H	1800.0	(0)	1534.0	(2)	1649.1	(1)	1151.4	(2)	1797.6	(1)		
		Average (Sum)				1332.3	(13)	1084.6	(19)	1406.4	(10)	1119.5	(17)	1473.5	(11)
		200	200	d_S	c_L	582.8	(4)	393.1	(4)	730.5	(3)	367.5	(4)	1595.9	(2)
c_H	1800.0				(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
d_L	c_L			1574.1	(1)	1659.7	(1)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
	c_H			1800.0	(0)	1583.5	(1)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
240	d_S		c_L	220.1	(5)	44.5	(5)	1083.7	(2)	484.8	(4)	1144.1	(2)		
			c_H	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
	d_L		c_L	1800.0	(0)	1800.0	(0)	1800.0	(0)	1425.1	(2)	1800.0	(0)		
			c_H	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)	1800.0	(0)		
	Average (Sum)				1422.1	(10)	1360.1	(11)	1576.8	(5)	1409.7	(10)	1692.5	(4)	
	Total: Average (Sum)				935.2	(85)	792.9	(97)	1022.8	(73)	861.0	(87)	1030.6	(76)	
Average Exit gap [%]				20.8		5.3		9.5		5.4		9.5			

Table 3: Number 'opt' of instances solved to optimality (from a total of five instances per subset), and average solution times t in seconds. For the sake of completeness, also the results of M3 from [17] and the average exit gaps are displayed.

ideas presented in Sect. 3 lead to significant reductions of the integer programs associated with M1 and M2. To be more precise, while the savings in the number of constraints is about 60% in both cases (compared to M3), in the case of the number of variables it ranges from about 32% (for M1*) to 45% (for M2*). These reductions become even more remarkable when referring only to the comparison of the literature version M1 (resp. M2) with the version M1* (resp. M2*) improved in the context of this work. On average, here we end up with roughly 75% fewer variables (in both cases) and, depending on the formulation, between 67% and 80% fewer constraints. While for a fixed number of items n and a fixed model we previously saw very little variation in the indicators n_{var} and n_{con} among the several groups of instances, we now observe that our reductions are particularly successful when short and/or low-resource jobs are considered (see d_S and c_L). On the one hand, these constellations tend to lead to particularly few interactions between the jobs and therefore allow for better heuristic solutions (typically leading to a small value of k^*), which is also clearly supported by the results from Tab. 1. On the other hand, for the case c_H , the number of forbidden item combinations increases significantly, so that, for

example, a sometimes substantial number of valid inequalities (see Reduction (b) in Sect. 3) must be added to the model.

In a next step, we focus on the performance of $M1^*$ and $M2^*$ when addressing the exact solution of the benchmark instances. To this end, we tabulate the obtained results in Tab. 3 and compare them with the previous state of the literature (that is, $M1$, $M2$, and $M3$ from [17]). First, we note that the contributions from Sect. 3 helped to significantly increase the number of instances solved to optimality. More specifically, the modifications to $M1$ ($M2$, resp.) resulted in 12 (14, resp.) additional proven optimal solutions, so that both formulations now perform considerably better than $M3$ from [17].

Remark 4.2. Interestingly, in at least seven additional cases $M2^*$ already had the correct optimal value, but failed to prove the optimality within the given time limit.

Due to its small model size and the fact that, for example, the reduction related to K is particularly promising for large values of n , $M3$ is still the best formulation for the very small instances with $n = 50$ items, but loses this leading position more and more clearly for larger instances (especially in comparison with $M1^*$). Overall, it is noticeable in Tab. 3 that the performance of $M1^*$ and $M2^*$ has improved, especially for many instances from the constellation (d_S, c_L) , which further supports the observations made in Tab. 2 that the reductions are particularly strong for these cases. However, the ideas from Sect. 3 not only contribute to an overall improvement in the number of optimally solved instances, but (in many cases) also to considerably lower computation times required. On the one hand, this can be seen from the average values in Tab. 3, but it becomes somewhat more evident if we look at the percentage of optimally solved instances over time, see Fig. 2. Therein, it is clearly visible that at any point in time one of the models $M1^*$ or $M2^*$ possesses the most convincing performance. The generally smaller model size of $M2^*$ causes it to dominate $M1^*$ within the first two minutes, while in the long term $M1^*$ offers the better numerical results. We attribute this to the fact that the numerous coupling conditions in $M1^*$ (i.e., the implications inherent to Constraints (3)–(5)) have a large effect in the deeper layers of the branch-and-bound tree, since already the specification of a small set of variables actually fixes a much larger set of variables to integer values. Moreover, the methods contained in Reduction (a), and the fact that the valid cuts from Reduction (b) can be formulated in a stronger way may also have contributed to the slightly better overall performance of $M1^*$.

While these considerations refer only to the successfully solved instances, the exit gaps provided in the last row of Tab. 3 also indicate the improvements with respect to all instances. Moreover, Fig. 3 additionally gives an overview of the average objective values (but only for the 80 instances with $n \geq 150$) over time. Besides the obvious and substantial improvements of $M1^*$ and $M2^*$ (over the original versions $M1$ and $M2$), we highlight the very good performance of $M2^*$ at all instants of time, which we again particularly attribute to the much smaller model size. Although Fig. 3 might suggest this, $M2^*$ is nevertheless not better than $M1^*$ for every single instance. More precisely, $M2^*$ was able to beat $M1^*$ in terms of the objective value in 26 cases, while the reverse situation occurred in 13 cases. Consequently, both formulations have proven beneficial for the instances considered here. From a general point of view, however, it can be said that the advantages of $M1^*$ lie in particular in its ability to obtain proven optimal solutions, while $M2^*$ is very well suited to provide reasonably good approximate solutions in a short time even for challenging instances of the TBPP-FU. The latter is additionally supported by the fact that the best objective value achieved within the time limit (across all instances with $n \geq 150$ and all models) averaged 38.45, and thus $M2^*$ was typically very close to the actual optimal value after only about two minutes according to Fig. 3. This property also emphasizes the applicability of $M2^*$ in some practically-oriented scenarios.

Overall, it can be concluded that the methods presented in Sect. 3 not only substantially improve the

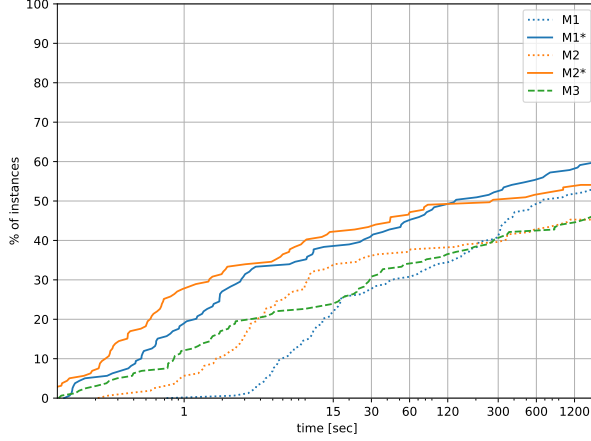


Figure 2: Temporal development of the number of instances solved to optimality by the various formulations.

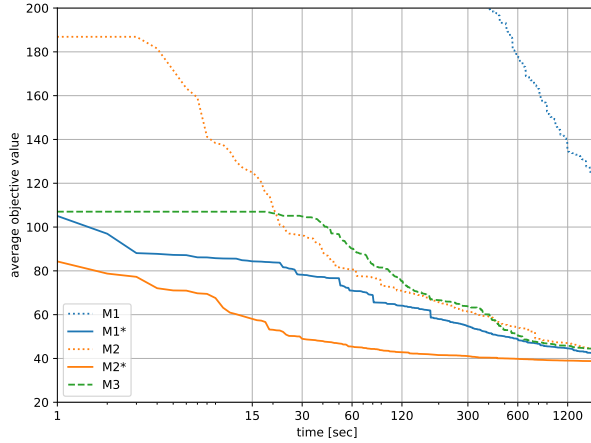


Figure 3: Temporal development of the average objective value for $n \geq 150$ by the various formulations. For completeness, we mention that the line of M1 starts at about 350 on the vertical axis.

models M1 and M2 individually, but also result in the advantageous features of the alternative approach M3 listed in [17] now being barely discernible in the numerical comparison. As a consequence, M3 is outperformed in almost every respect by M1* and M2*.

5 Conclusions

In this article, we dealt with the temporal bin packing problem with fire-ups, a relatively new decision making problem in operations research typically leading to integer models of challenging size. Even though some fundamental methods for obtaining more tractable formulations have already been described in the recent literature, these investigations do not yet turn out to be “complete” upon closer inspection, especially because the incorporation of heuristic information has so far only been possible for a few special cases. Therefore, the contributions of this article aim in particular at three methods to improve existing ILP models: “optimizing” the set of constraints (by removing redundancies and tightening some inequalities), adding valid inequalities, and reducing the number of servers to be considered (thus considerably decreasing the overall model size). Based on numerical computations, the positive effects of the new techniques could be manifested. We underline not only the fact that, as a result of the improvements, a fixed model was able to solve up to 14 additional instances (compared to the previous

version from [17]) of the challenging benchmark set from [3], but also highlight that, in particular, the optimal solution of seven instances (five each by M1^{*} and M2^{*}, and three by both models) was obtained for the first time. Now that the investigation of assignment models for the TBPP-FU is somewhat “complete”, future research should focus in particular on flow-based models or branch-and-price approaches. Moreover, theoretical results dealing with the worst-case performance of heuristics for the TBPP-FU have not yet been addressed at all in the literature.

Declarations

Funding: Not applicable. **Conflicts of interest:** The authors declare that they do not have any conflicts of interest. **Availability of data and material:** The instances used in that paper were originally designed in [3] and can be found online, see <https://github.com/sibirbil/TemporalBinPacking>. **Code availability:** The instances were solved by the commercial software Gurobi. The underlying implementation of the models in Python can be found in <https://github.com/wotzlauff/tbpp-cf2>.

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